

Explain how perceptron learning algorithm works for an AND Gate. ①
Consider $w_1 = 0.19$ $w_2 = 0.19$.

Learning rule as:

$$w = w + \lambda(a_k - d_k)$$

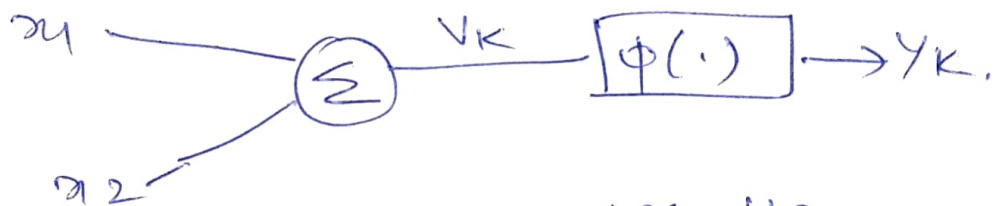
where λ - learning rate = 0.5

a_k - Actual output

d_k - desired output.

Truth Table :-

x_1	x_2	y
0	0	0
0	1	0
1	0	0
1	1	1



$$v_k = x_1 w_1 + x_2 w_2$$

$$y_k = \phi(v_k)$$

ϕ - Threshold (Heaviside) function

$$y_k = \begin{cases} 1 & v_k \geq 0 \\ 0 & v_k < 0 \end{cases}$$

Q. ... Problem:
1st Instance:-

(2)

$$x_1 = 0 \quad x_2 = 0$$

$$v_k = (0 * 0.19) + (0 * 0.19) = 0.$$

$$y_k = 1 \text{ as } v_k \geq 0.$$

Update the weights

$$W = W + \lambda(a_k - d_k)$$

$$W = W + 0.5(0 - 1)$$

$$w_1 = 0.19 + 0.5(-1) = \cancel{0.34} - 0.31$$

$$w_2 = 0.19 + 0.5(-1) = \cancel{0.4} - 0.31$$

Use the updated weights for 2nd Instance.

$$x_1 = 0 ; x_2 = 1$$

$$w_1 = -0.31 ; w_2 = -0.31$$

$$v_k = (0 * -0.31) + (1 * -0.31)$$

$$v_k = -0.31$$

$$y_k = 0 \text{ as } v_k < 0.$$

$\therefore$ ~~W =~~ ~~W +~~ $a_k = d_k$ so no
update of weights.

Continue the process.

Sample Problems:

(3)

- ① Design a McCulloch Pitts model for a OR Gate. Assume data wherever necessary.
- ② Design a perceptron model for a OR Gate without bias. Assume data wherever necessary.
- ③ Design a OR Gate for the combination of three variables x_1, x_2 , and x_3 with Bias as 1.

Weights as

$$W_0 = 0.4$$
$$W_1 = 0.4$$
$$W_2 = 0.4$$
$$W_3 = 0.4$$